

What is Data Science

The term “data science” combines two key elements: “data” and “science.”

Data: It refers to the raw information that is collected, stored, and processed. In today’s digital age, enormous amounts of data are generated from various sources such as sensors, social media, transactions, and more. This data can come in structured formats (e.g., databases) or unstructured formats (e.g., text, images, videos).

Science: It refers to the systematic study and investigation of phenomena using scientific methods and principles. Science involves forming hypotheses, conducting experiments, analyzing data, and drawing conclusions based on evidence.

“Data science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data. In simpler terms, data science is about obtaining, processing, and analyzing data to gain insights for many purposes.”

Need of Data Science

- Data science may detect patterns in seemingly unstructured or unconnected data, allowing conclusions and predictions to be made.
- Tech businesses that acquire user data can utilize strategies to transform that data into valuable or profitable information.
- Data Science can extract valuable insights from any type of available information, and this skill set is what enables them to do so.
- Make data scientists more productive by helping them accelerate and deliver models faster, and with less error.
- Better decision-making: The insights gleaned from the analyzed data can help organizations make informed decisions that meet the needs of the existing problem and not just infer a solution without basic validation.
- Performance measurement and increased efficiency: Data generated from different sources can be used as a measuring tool, allowing companies to leverage data to measure growth and detect pitfalls to prepare for and mitigate them easily.
- Prevent future risks: Through data science methods such as predictive analysis, you can use your data to highlight areas of potential risk.

Big data and Data Science

What is Big Data?

Big Data refers to the vast volumes of data generated at high velocity from a variety of sources. This data is characterized by the three V's: Volume, Velocity, and Variety.

Volume: Big Data involves large datasets that are too complex for traditional data processing tools to handle. These datasets can range from terabytes to petabytes of information.

Velocity: Big Data is generated in real-time or near real-time, requiring fast processing to extract meaningful insights.

Variety: The data comes in multiple forms, including structured data (like databases), semi-structured data (like XML files), and unstructured data (like text, images, and videos).

Big data and Data Science

Data Science is an interdisciplinary field that utilizes scientific methods, algorithms, and systems to extract knowledge and insights from structured and unstructured data. It encompasses a variety of techniques from statistics, machine learning, data mining, and big data analytics.

Data Scientists use their expertise to:

Analyze: They examine complex datasets to identify patterns, trends, and correlations.

Model: Using statistical models and machine learning algorithms, they create predictive models that can forecast future trends or behaviors.

Interpret: They translate data findings into actionable business strategies and decisions.

Big data and Data Science

Aspect	Big Data	Data Science
Definition	Handling and processing vast amounts of data	Extracting insights and knowledge from data
Objective	Efficient storage, processing, and management of data	Analyzing data to inform decisions and predict trends
Focus	Volume, velocity, and variety of data	Analytical methods, models, and algorithms
Primary Tasks	Collection, storage, and processing of data	Data analysis, modeling, and interpretation
Tools/Technologies	Hadoop , Spark , NoSQL databases (e.g., MongoDB)	Python , R, TensorFlow , Scikit-Learn
Data Types	Structured, semi-structured, and unstructured data	Processed and cleaned data for analysis
Outcome	Accessible data repositories for analysis	Actionable insights, predictive models
Skill Set	Data engineering, distributed computing	Statistical analysis, machine learning, programming
Typical Roles	Data Engineers, Big Data Analysts	Data Scientists, Machine Learning Engineers
Applications	Real-time data processing, large-scale data storage	Predictive analytics, data-driven decision making
Key Techniques	Distributed computing, data warehousing	Statistical modeling , machine learning algorithms

Difference between Structured and Unstructured data

Structured data –

Structured data is data whose elements are addressable for effective analysis. It has been organized into a formatted repository that is typically a database.

It concerns all data which can be stored in database SQL in a table with rows and columns. They have relational keys and can easily be mapped into pre-designed fields. Today, those data are most processed in the development and simplest way to manage information.

Example: Relational data.

Unstructured data –

Unstructured data is a data which is not organized in a predefined manner or does not have a predefined data model, thus it is not a good fit for a mainstream relational database. So for Unstructured data, there are alternative platforms for storing and managing, it is increasingly prevalent in IT systems and is used by organizations in a variety of business intelligence and analytics applications.

Example: Word, PDF, Text, Media logs.

Quantitative data and Categorical data

Quantitative data:

Quantitative data consists of numerical insights that represent quantities or measurements. It provides insights into the magnitude, size, or amount of something. Quantitative data can be further classified into discrete and continuous data.

Categorical data:

Categorical data, also known as qualitative or nominal data, comprises information that can be grouped into distinct categories or labels. It conveys descriptive details about the qualitative attributes or characteristics of an object, individual, or event. Common examples include colors, genders, and types of cars, or simple yes or no responses.

Quantitative data Vs Categorical data

	Categorical Data	Quantitative Data
Type of Values	Labels or categories	Numerical values
Nature	Qualitative	Quantitative
Mathematical Operations	Limited (frequencies, proportions)	Extensive (mathematical calculations, statistical analysis)
Level of Measurement	Nominal or ordinal	Interval or ratio
Examples	Colors, genders, car types	Height, weight, temperature

Big Data Vs Little/Small Data

Big Data:

It can be represented as large chunks of structured and unstructured data. The amount of data stored is immense. It is therefore important for analysts to thoroughly dig the whole thing into making it relevant and useful to make proper business decisions.

In short, datasets that are really huge and complex that conventional data processing techniques can not manage them are known as big data.

Little/Small Data:

It can be defined as small datasets that are capable of impacting decisions in the present. Anything that is currently ongoing and whose data can be accumulated in an Excel file. Small Data is also helpful in making decisions, but does not aim to impact the business to a great extent, rather for a short span of Small data can be described as small datasets that are capable of having an influence on current decisions. Small data is also useful in decision-making but is not intended to have a large impact on business, rather for a short period of time.

In nutshell, data that is simple enough to be used for human understanding in such a volume and structure that makes it accessible, concise, and workable is known as small data.

Data Science Process /Life Cycle

Some steps are necessary for any of the tasks that are being done in the field of data science to derive any fruitful results from the data at hand.

Data Collection – After formulating any problem statement the main task is to calculate data that can help us in our analysis and manipulation. Sometimes data is collected by performing some kind of survey and there are times when it is done by performing scrapping.

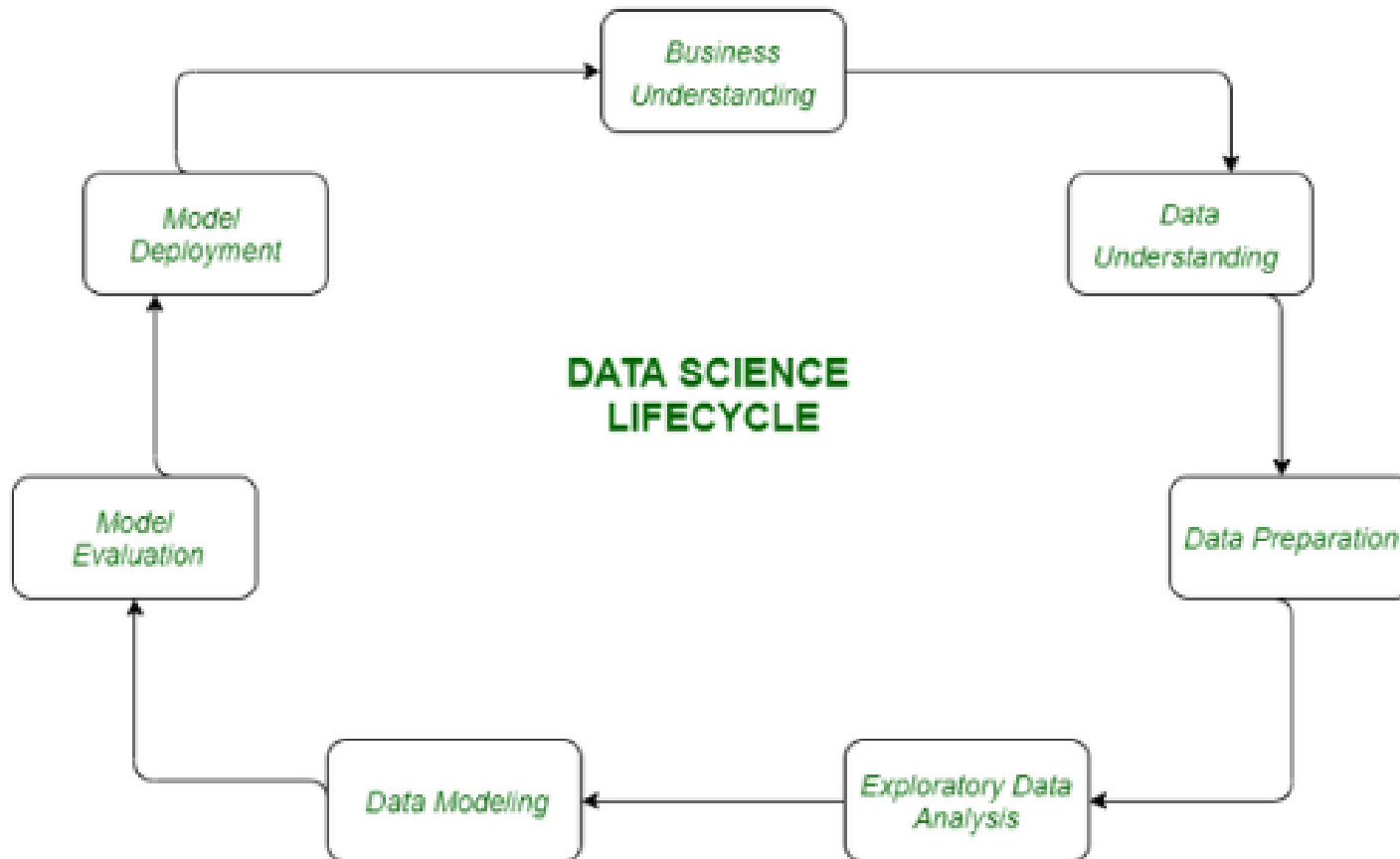
Data Cleaning – Most of the real-world data is not structured and requires cleaning and conversion into structured data before it can be used for any analysis or modeling.

Exploratory Data Analysis – This is the step in which we try to find the hidden patterns in the data at hand. Also, we try to analyze different factors which affect the target variable and the extent to which it does so. How the independent features are related to each other and what can be done to achieve the desired results all these answers can be extracted from this process as well. This also gives us a direction in which we should work to get started with the modeling process.

Model Building – Different types of machine learning algorithms as well as techniques have been developed which can easily identify complex patterns in the data which will be a very tedious task to be done by a human.

Model Deployment – After a model is developed and gives better results on the holdout or the real-world dataset then we deploy it and monitor its performance. This is the main part where we use our learning from the data to be applied in real-world applications and use cases.

Data Science Process /Life Cycle



Thank you